

Project Design Phase
Proposed Solution

Date	29 June 2026
Team ID	SWTID-2026-2260
Project Name	Smart Lender
Maximum Marks	2 Marks

Proposed Solution:

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Banks and financial institutions face challenges in manually evaluating loan applications, resulting in delays, human errors, and inconsistent credit assessment. An intelligent system is required to automate the loan approval process and improve decision-making accuracy.
2.	Idea / Solution description	Smart Lender is a Machine Learning-based web application that predicts loan approval by analyzing applicant information such as gender, marital status, education, employment status, income, loan amount, loan term, credit history, and property area. The application uses multiple classification algorithms and deploys

		the best-performing XGBoost model through a Flask web application on IBM Cloud for real-time predictions.
3.	Novelty / Uniqueness	The proposed solution compares multiple machine learning algorithms—Decision Tree, Random Forest, K-Nearest Neighbors (KNN), and XGBoost—and selects the most accurate model for deployment. It provides quick, reliable, and automated loan approval predictions through a simple web interface, reducing manual intervention and improving operational efficiency.
4.	Social Impact / Customer Satisfaction	The solution helps financial institutions process loan applications faster, reduces the risk of loan defaults, and improves the accuracy of lending decisions. Customers benefit from quicker responses, greater transparency, and an improved banking experience, while banks achieve better customer satisfaction and operational efficiency.
5.	Business Model (Revenue Model)	The Smart Lender application can be offered as a Software-as-a-Service (SaaS) platform to banks, financial institutions, and NBFCs through subscription-based licensing. Additional revenue can be generated through cloud hosting, enterprise customization, maintenance services, and technical support.

6.	Scalability of the Solution	The application is designed with a scalable architecture using Flask and IBM cloud, allowing it to support a large number of users and loan applications. The system can be easily enhanced by integrating new machine learning models, additional applicant features, mobile applications, and advanced analytics in the future.
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